

# Technical Document #448: Deep Learning - Comprehensive Overview

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Transformer architecture relies entirely on self-attention mechanisms, dispensing with recurrence and convolutions. It has become the foundation for modern NLP models.

Long short-term memory (LSTM) networks are a type of RNN that can learn long-term dependencies. They use gating mechanisms to control the flow of information.

Transfer learning is a technique where a model trained on one task is re-purposed on a second related task. It has revolutionized computer vision and NLP.

The rapid advancement of technology has transformed how organizations approach problem-solving and innovation. Companies must adapt to stay competitive in an ever-evolving landscape.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

## Key Takeaways:

- Dropout is a regularization technique where randomly selected neurons are ignored during training.
- Generative adversarial networks (GANs) consist of two neural networks competing against each other.
- Transfer learning is a technique where a model trained on one task is re-purposed on a second related task.